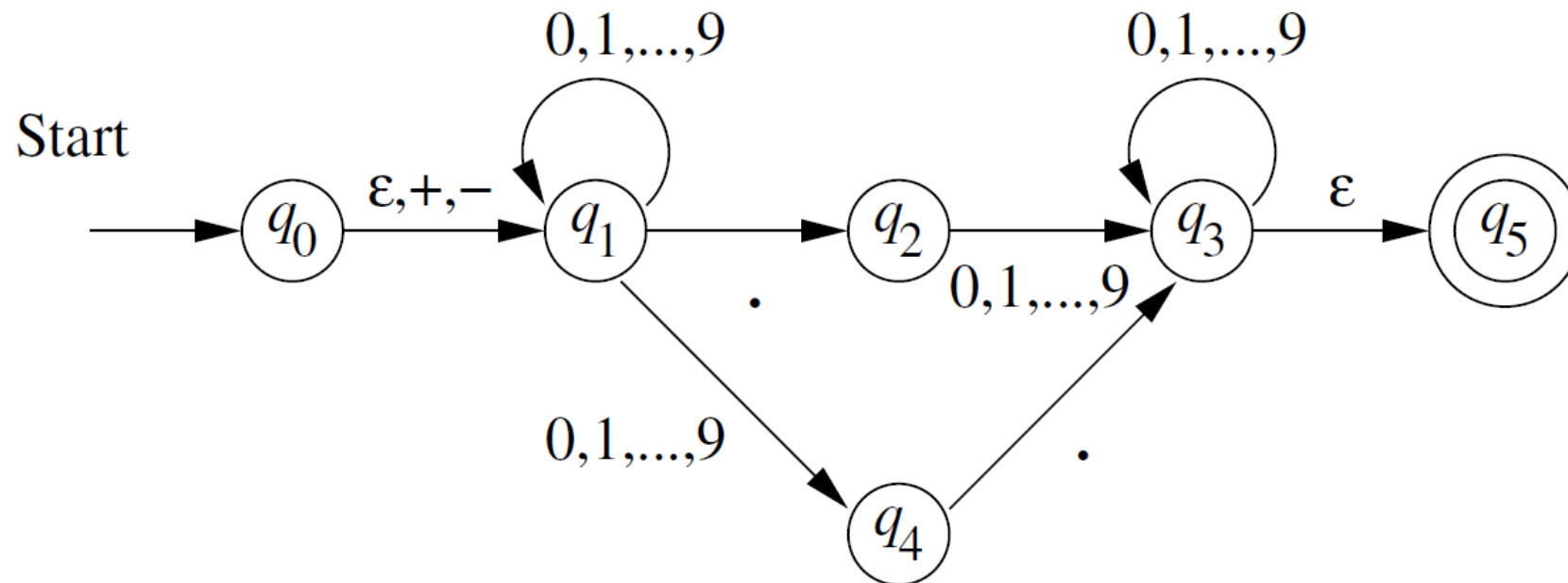


Automata and Formal Languages

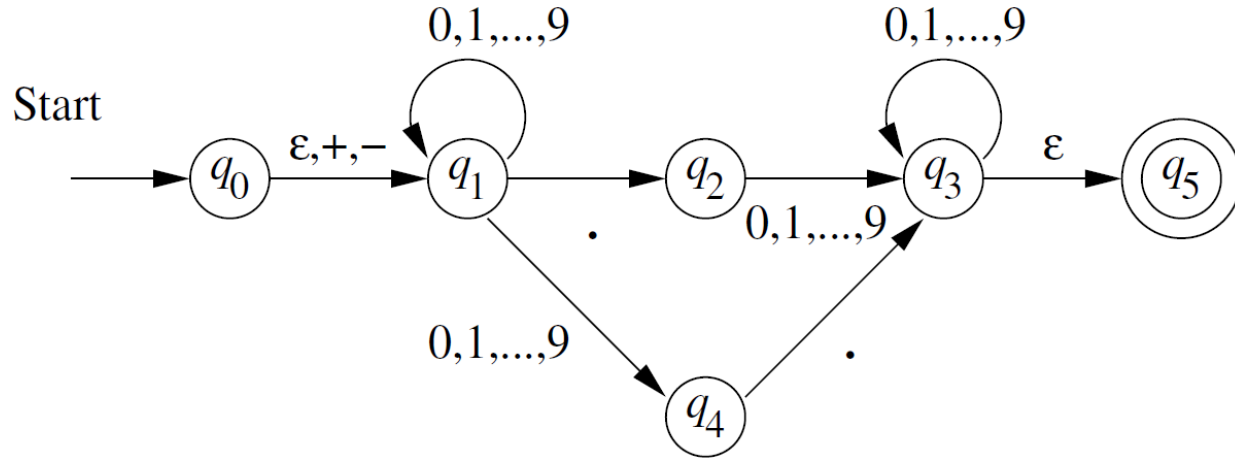
Dr. Sankhadeep Chatterjee

NFA with ϵ Transition

- NFA where ϵ -transitions are allowed
- Only difference with NFA: $\delta: Q \times (\Sigma \cup \epsilon) \rightarrow P(Q)$
- Example: ϵ -NFA that accepts decimal numbers



NFA with ϵ Transition



Transition Diagram

	ϵ	$+, -$	$.$	$0, 1, \dots, 9$
q_0	$\{q_1\}$	$\{q_1\}$	$\emptyset$	$\emptyset$
q_1	$\emptyset$	$\emptyset$	$\{q_2\}$	$\{q_1, q_4\}$
q_2	$\emptyset$	$\emptyset$	$\emptyset$	$\{q_3\}$
q_3	$\{q_5\}$	$\emptyset$	$\emptyset$	$\{q_3\}$
q_4	$\emptyset$	$\emptyset$	$\{q_3\}$	$\emptyset$
q_5	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

Transition Table

ϵ -closure

Find ϵ -close of state q by following all transitions out of q with label ϵ

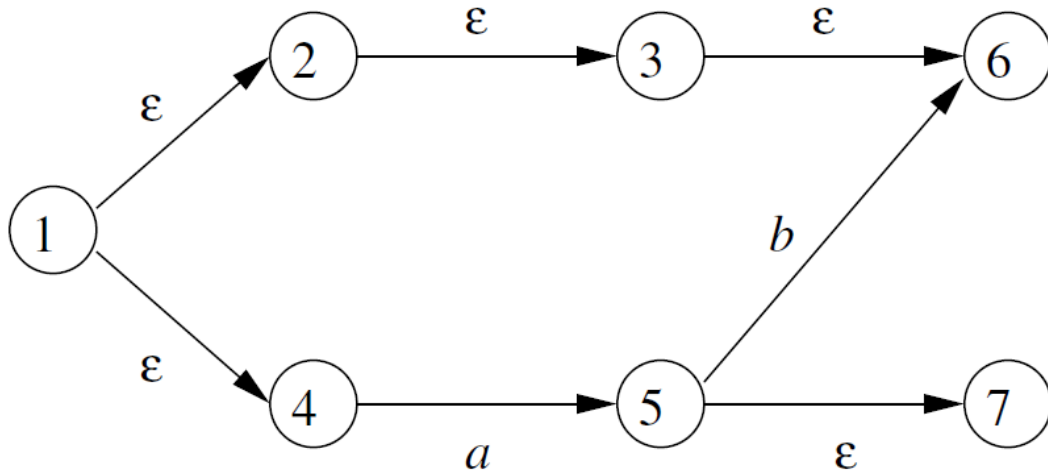
ECLOSE(q):

$E = \{q\}$

For all p in $\delta(q, \epsilon)$

$E \leftarrow E \cup \text{ECLOSE}(p)$

Return E



$\text{ECLOSE}(1) = \{1, 2, 3, 6, 4\}$

Transition Function of ϵ -NFA

- Consider $w = xa$ where a is the last symbol and x is a rest of w .
- Assume $\hat{\delta}(q, x) = \{p_1, p_2, \dots, p_k\}$

If

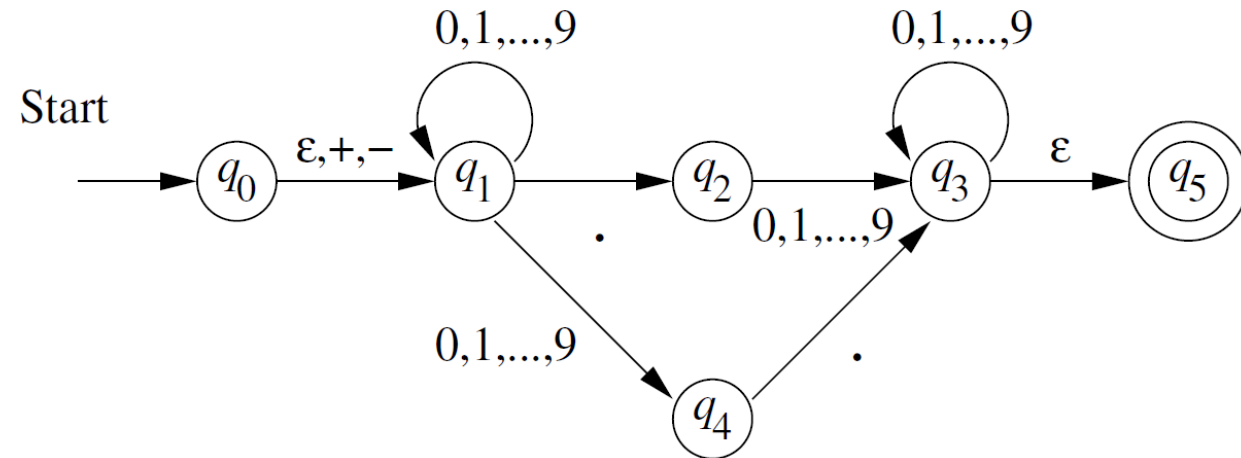
$$\bigcup_{i=1}^k \delta(p_i, a) = \{r_1, r_2, \dots, r_m\}$$

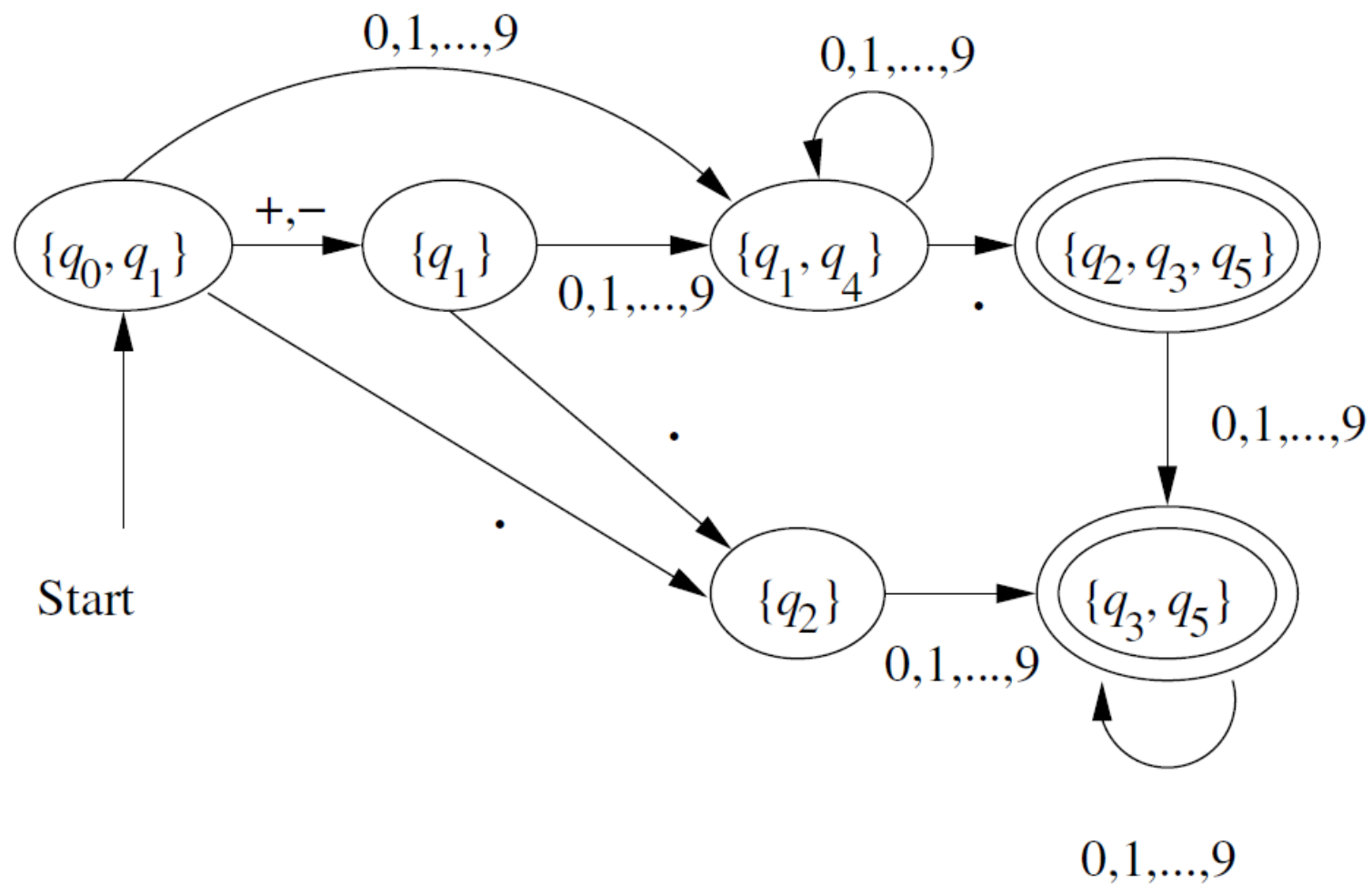
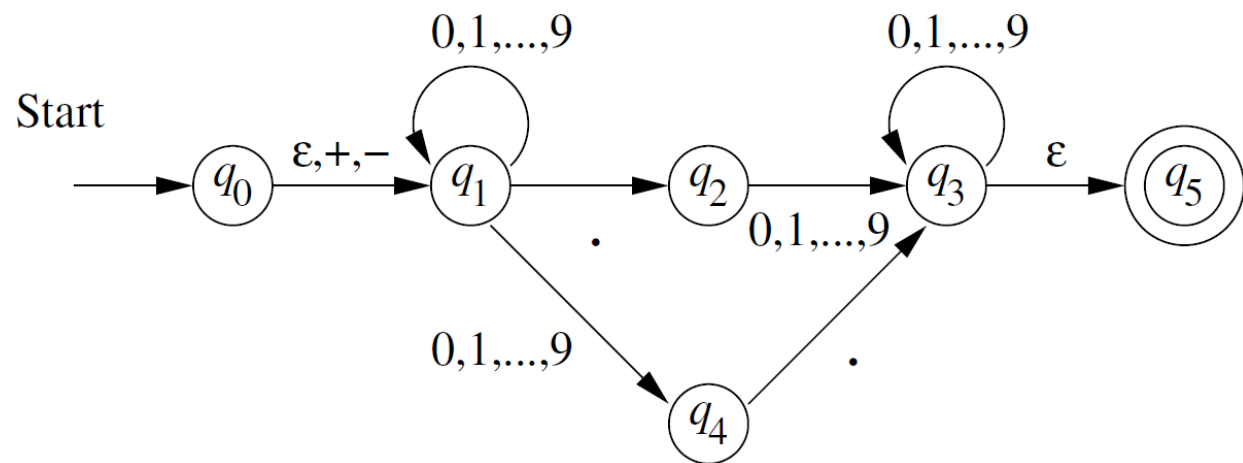
then

$$\hat{\delta}(q, w) = ECLOSE\{r_1, r_2, \dots, r_m\}$$

Eliminating ϵ -Transition

- Consider ϵ -NFA $E = (Q_E, \Sigma, \delta_E, q_0, F_E)$
- Consider DFA $D = (Q_D, \Sigma, \delta_D, q_D, F_D)$
- Conversion:
 - $Q_D = \{S | S \subseteq Q_E \text{ and } \epsilon - \text{closed}\}$
 - Σ remains same
 - $\delta_D(S, a)$ is computed as follows:
 - Assume $S = \{p_1, p_2, \dots, p_k\}$
 - $\bigcup_{i=1}^k \delta_E(p_i, a) = \{r_1, r_2, \dots, r_m\}$
 - $\delta_D(S, a) = ECLOSE(\{r_1, r_2, \dots, r_m\})$
 - $q_D = ECLOSE(q_0)$
 - $F_D = \{S | S \subseteq Q_D \text{ and } S \cap F_E \neq \emptyset\}$



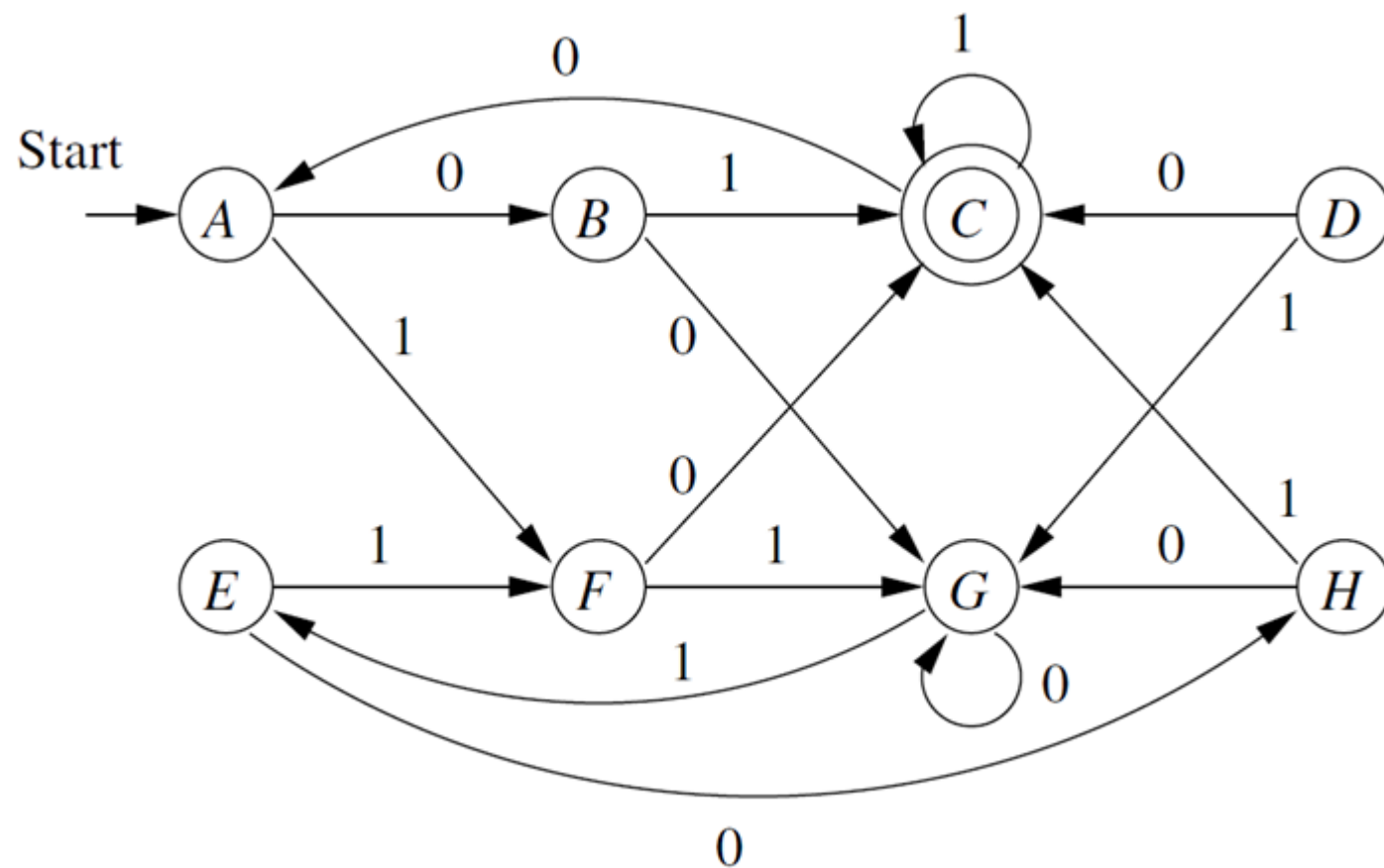


Distinguishable States of DFA

1. If p is an **accepting** state and q is **nonaccepting**, then the pair $\{p, q\}$ is **distinguishable**.
2. Let p and q be states such that for some input symbol a ,
$$r = \delta(p, a) \text{ and } s = \delta(q, a)$$

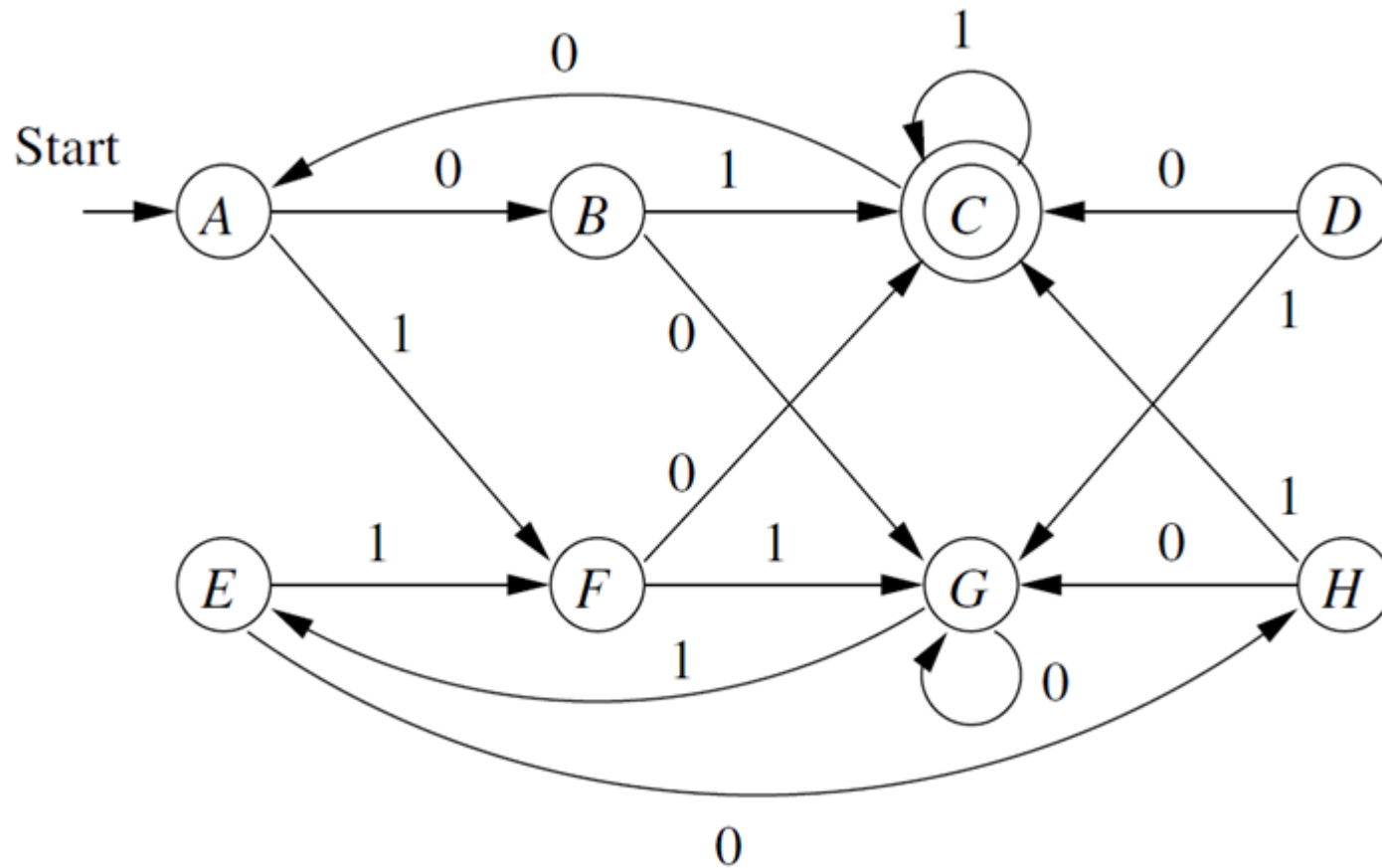
If r and s are a pair of states known to be **distinguishable** then, $\{p, q\}$ is **distinguishable**.

Table Filling Algorithm



<i>B</i>	<i>x</i>						
<i>C</i>	<i>x</i>	<i>x</i>					
<i>D</i>	<i>x</i>	<i>x</i>	<i>x</i>				
<i>E</i>		<i>x</i>	<i>x</i>	<i>x</i>			
<i>F</i>	<i>x</i>	<i>x</i>	<i>x</i>		<i>x</i>		
<i>G</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	
<i>H</i>	<i>x</i>		<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>	<i>x</i>
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>

DFA Minimization



- After partitioning the states into equivalent blocks we get:
 $(\{A, E\}, \{B, H\}, \{C\}, \{D, F\}, \{G\})$

DFA Minimization

